11.15 Classwork: Periodic Function Modeling (calculator) — Markscheme

1. [Maximum mark: 7]

   A particle moves along a straight line so that its velocity, v m s^{-1}, after t seconds is given by
   $$v(t) = e^{\sin t} + 4\sin t, \text{ for } 0 \le t \le 6.$$

   (a) Find the value of t when the particle is at rest. [2]

   Recognising at rest $v = 0$. **M1**

   GDC: solve $e^{\sin t} + 4\sin t = 0$ *on* $[0, 6]$.

   $t = 3.34692\ldots = 3.35$ (seconds) **A1**

   Note: Award M1A0 for additional solutions to $v = 0$ *(e.g.* $t = -0.205$ *or* $t = 6.08$*).*

   (b) Find the acceleration of the particle when it changes direction. [3]

   Recognising particle changes direction when $v = 0$ OR when $t = 3.347\ldots$ **M1**

   GDC: evaluate $\dfrac{dv}{dt}$ *at* $t = 3.347\ldots$.

   $a = -4.71439\ldots$

   $a = -4.71$ (m s^{-2}) **A2**

   (c) Find the total distance travelled by the particle. [2]

   distance travelled $= \displaystyle\int_0^6 |v(t)|\,dt$ **A1**

   GDC: split at $t = 3.347$ *since* v *changes sign there.*

   $$\int_0^{3.347} v(t)\,dt - \int_{3.347}^{6} v(t)\,dt = 14.310\ldots + 6.443\ldots = 20.753\ldots$$

   $= 20.8$ (metres) **A1**

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2. [Maximum mark: 13]

   $H(t) = a\cos(7.8\,t) + b, \quad 0 \le t \le 10.$ Min at 1 m, max at 1.8 m.

   (a) Find the period of the function. [2]

   $7.8 = \dfrac{2\pi}{\text{period}}$ **M1** period $= \dfrac{2\pi}{7.8} = \dfrac{10\pi}{39} \approx 0.806$ (seconds) **A1**

(b) Find (i) a; (ii) b. [3]

METHOD 1. Amplitude $= \dfrac{\max - \min}{2} = \dfrac{1.8-1}{2} = 0.4$. **M1**

The weight is released at $t=0$ at the minimum, and $\cos(0)=1$, so $H(0)=a+b=1$ — therefore $a<0$.

(i) $a=-0.4$ **A1**

(ii) $b = \dfrac{\max + \min}{2} = \dfrac{1.8+1}{2} = 1.4$ **A1**

METHOD 2. Form two simultaneous equations: $H(0)=a+b=1$ and $H(\pi/7.8) = -a+b = 1.8$. **M1**

$a=-0.4$, $b=1.4$ **A1A1**

(c) Find the number of times the weight reaches its maximum height in the first 5 seconds. [2]

EITHER $\dfrac{5}{\text{period}} = \dfrac{5}{0.8055\ldots} = 6.207\ldots < 6\frac{1}{2}$ **(A1)**

OR maximums occur at $t = 0.403, 1.21, 2.01, 2.82, 3.62, 4.43$ **(A1)**

THEN 6 times **A1**

(d) Find the first time the weight reaches a height of 1.5 metres. [2]

Recognising $H(t)=1.5$. **M1**

$-0.4\cos(7.8t)+1.4 = 1.5 \;\Rightarrow\; \cos(7.8t) = -0.25$

GDC: solve on $[0,10]$.

$t = 0.234$ (seconds) **A1**

(e) Find the probability the height is > 1.5 metres at the picture time. [4]

Find the second time $H=1.5$ in the first period: $t=0.572$. **M1**

In each period, $H>1.5$ for $0.572-0.234 = 0.338$ seconds. **(A1)**

Equivalently: total time > 1.5 *in* $[0,5]$ *is* ≈ 2.028 *seconds (six full crossings).*

Multiply by 6 and divide by 5: **M1**

$P(H>1.5) = \dfrac{0.338\times 6}{5} = 0.4055\ldots = 0.406$ **A1**

3. [Maximum mark: 13]

$H(t) = a\sin(b(t-c)) + d$. First high tide 04:30, period 12 hours, $H \in [2.2, 6.8]$.

(a) Show that $b = \frac{\pi}{6}$. [1]

$12 = \frac{2\pi}{b}$, so $b = \frac{2\pi}{12} = \frac{\pi}{6}$. **A1 AG**

(b) Find a. [2]

$a = \frac{\max - \min}{2} = \frac{6.8 - 2.2}{2}$ **M1**

$= 2.3$ (m) **A1**

(c) Find d. [2]

$d = \frac{\max + \min}{2} = \frac{6.8 + 2.2}{2}$ **M1**

$= 4.5$ (m) **A1**

(d) Find the smallest possible value of c. [3]

METHOD 1. Substitute $t = 4.5$, $H = 6.8$: **(A1)**

$6.8 = 2.3\sin\left(\frac{\pi}{6}(4.5 - c)\right) + 4.5 \;\Rightarrow\; \sin\left(\frac{\pi}{6}(4.5 - c)\right) = 1$

M1

$\frac{\pi}{6}(4.5 - c) = \frac{\pi}{2} \;\Rightarrow\; 4.5 - c = 3 \;\Rightarrow\; c = 1.5$ **A1**

METHOD 2. Standard sine peaks one quarter-period after the inflection at $t = c$. Quarter-period $= 3$, so $4.5 - c = 3$. **(A1)(M1)**

$c = 1.5$ **A1**

(e) Find the height of the water at 12:00. [2]

Substitute $t = 12$ (graphically or algebraically): **M1**

$H(12) = 2.3\sin\left(\frac{\pi}{6}(12 - 1.5)\right) + 4.5 = 2.3\sin(5.4977\ldots) + 4.5 = 2.87$ (m) **A1**

(f) Number of hours over a 24-hour period when tide is higher than 5 m. [3]

Solve $5 = 2.3\sin\left(\frac{\pi}{6}(t - 1.5)\right) + 4.5$. **M1**

GDC: $t = 1.919, 7.082$ *in* $[0, 12]$, *then* $t = 13.919, 19.082$ *in* $[12, 24]$. **(A1)**

Total: $2(7.082 - 1.919) = 10.326$. **A1**

≈ 10.3 hours.

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4. [Maximum mark: 13]

(a) Find the height of point C above the ground. [2]

$\tan(0.6) = \dfrac{h}{12}$ **M1**

$h = 12\tan(0.6) = 8.20964\ldots = 8.21$ (m) **A1**

(b) Find the angle of elevation of point C from B. [2]

Distance from B to base $= 12 - 7 = 5$ m. **(A1)**

$\tan(\angle B) = \dfrac{8.21}{5} = 1.6419\ldots \Rightarrow \angle B = 1.024$ (rad) **A1**

Accept 58.7.

(c) Find the length of each blade. [2]

Let x be the blade length: $x + 1.8 + 2.5 = 8.21$ **(A1)**

$x = 3.91$ (m) **A1**

(d) Find p and q in $h(t) = p\cos\left(\dfrac{3\pi}{10}t\right) + q$. [4]

METHOD 1.

Blade length = amplitude: $p = \dfrac{\max - \min}{2}$ **M1**

$p = 3.91$ **A1**

Centre of windmill = vertical shift: $q = \dfrac{\max + \min}{2}$ **M1**

$q = 8.21$ **A1**

METHOD 2. Two equations from max and min positions: **M1 M1**

$12.12 = p\cos(0) + q, \quad 4.30 = p\cos(\pi) + q$

$p = 3.91,\ q = 8.21$ **A1A1**

(e) Find n, the number of times D passes overhead in 1 minute. [3]

Period $= \dfrac{2\pi}{3\pi/10} = \dfrac{20}{3} \approx 6.667$ seconds. **M1**

Rotations per minute $= \dfrac{60}{\text{period}}$ **M1**

$n = 9$ (must be an integer) **A1**

Accept $n = 10$, $n = 18$, $n = 19$.